

# The Theory of Everything

A Programme Built on the Zero Point, the Observer and Information

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## Abstract

The framework of P. LeBaux is restated around a parameter-free zero point  $Z$ , an observer who adopts it as a relative centre, and a postulate that information can be created and destroyed. We show that a one-outcome source carries 0 Shannon bits, so the “two bits” the author reads from  $Z$  belong to the observer’s description, not to empty space; that creation and destruction of information are separate claims, neither implying the other; and where each is supported in physics (measurement, Landauer erasure) or disputed (black-hole evaporation). We give the elementary results on computational collapse, an analysis of general relativity, the status of the six open Millennium Problems, and an empirical test with an HSAM observer. **The claims that the remaining Clay problems and consciousness are solved are recorded as claims; no derivation of either appears here. This document presents a theory, a set of postulates and their stated consequences; it is not a rule or law of nature.** It is a programme statement, not a completed theory of everything.

**Keywords:** computational complexity; observer; correspondence principle; physical computation; HSAM.  
**MSC 2020:** 68Q15, 68Q17.

## Central equations

$$\sum_{k \geq 1} 2^{-k} = 1 \quad (\text{idealized collapse, Thm. A [21]}) \quad (1)$$

$$T(n) \geq \frac{2^n d}{v} \quad (\text{brute-force bound, Thm. B [17]}) \quad (2)$$

$$n_{\max} = \lfloor \log_2 M \rfloor \quad (\text{lookup bound, Thm. C [14]}) \quad (3)$$

$$\lim_{v, E \rightarrow \infty, d \rightarrow 0} T(n) = 0 \quad (\text{correspondence principle, §4 [30, 28]}) \quad (4)$$

$$H = - \sum_i p_i \log_2 p_i \quad (\text{Shannon entropy; } H = 0 \text{ for one outcome [2]}) \quad (5)$$

$$Q \geq k_B T \ln 2 \quad (\text{heat per erased bit [3, 4]}) \quad (6)$$

$$\log T = \log a + n \log b \quad (\text{exponential fit on hard 3-SAT [18, 19, 20]}) \quad (7)$$

Equations (1)–(3) are derived below; (5)–(6) are standard results; (4) is a postulate; (7) is the empirical test of §7.

## 1 Audit of the original monograph

The monograph (v1.0.0) contains nine claims. Proved: the counts  $2^N$  and  $\binom{N}{2}$ ; Euler’s identity (a standard theorem, independent of the framework). Definitional only:  $E = mc^2$  as written ( $m$  is defined as  $E/c^2$ ; it equals the invariant mass only in the zero-momentum frame). Not established: constant “crystallization”, nodal “firewalls” (the stated formula vanishes whenever a firewall exists), the field-only model of matter, the claimed proof of  $P = NP$  (it counts  $\binom{N}{2}$  planes, not  $2^N$  candidates, and sets  $\mathcal{T}_{\text{solve}} = 0$  while claiming  $O(N^2)$ ),

and the qualia postulate. The metric  $g_{ij} = J^T J$  is Riemannian, not Lorentzian. A full audit is given in the companion paper.

## 2 The zero point and information

**Axiom 1** ( $Z$ ). There is a zero point  $Z$  with no parameters, no vectors and no internal structure.

**Definition 1** (Observer). An observer is a physical system that adopts  $Z$  as its relative centre: it assigns itself coordinate 0 and measures all else relative to it.

**Proposition 1** (Relative centre). *Any point may serve as  $Z$  without changing physical laws that are translation-invariant.*

*Derivation.* If a law is invariant under  $x \mapsto x - p$  for every  $p$ , the choice of origin is conventional; by Noether's theorem this invariance corresponds to conservation of momentum [1].  $\square$

**Proposition 2** (Limit of the axiom).  *$Z$  alone entails no statement about number, geometry, fields, computation or the Millennium Problems.*

*Derivation.* A one-element structure satisfies  $Z$ . In it there is no second point, no metric and no operation, so any statement that fails in a one-element structure is not entailed by  $Z$ .  $\square$

Consequently every result below uses, beyond  $Z$ , (i) standard mathematics and (ii) the observer's own measured parameters (Definition 2). Which spacetime the observer inhabits is treated in §9.

**Proposition 3** (Information content of  $Z$ ). *A source with one possible outcome carries  $H = 0$  bits. A description of  $Z$  has Kolmogorov complexity  $K(Z) = O(1)$ , a constant depending on the description language.*

*Derivation.* With  $p = 1$ ,  $H = -1 \cdot \log_2 1 = 0$ . By the invariance theorem, complexity differs between universal languages by at most an additive constant [6].  $\square$

*Remark 1.* The “two bits” the author reads from  $Z$  (that it has no parameters, and that it can serve as a centre) are properties of the observer's description. The choice of origin is a gauge choice and carries no physical information (Prop. 1, [1]). Nothing is extracted from empty space; physical vacua are in any case not parameter-free (fields, metric, cosmological constant).

**Postulate 1** (Information, author's). Information can be created, and it can be destroyed.

**Status.** *Creation:* an observer gains  $H$  bits when a measurement outcome is not determined in advance; certified random bits exist [7], but they require devices, energy and a random seed, so this is gain by the observer, not creation from nothing. *Destruction:* erasing a bit dissipates at least  $k_B T \ln 2$  [3], confirmed experimentally [4], and avoidable by reversible computation [5]. *Fundamental loss:* Hawking proposed information loss in black-hole evaporation [8]; current calculations support preservation [9, 10]. *Inference:* “created, hence destroyed” is not a theorem; each claim must be established separately.

## 3 Devices and observers

**Definition 2** (Device). A device is  $D = (v, E, d, M, \epsilon)$ : maximum signal speed  $v \leq c$ ; energy budget  $E$ ; minimum distance  $d > 0$  moved per elementary step; memory capacity  $M$  bits; precision  $\epsilon > 0$ . An elementary step is a state change requiring motion of at least  $d$  at speed at most  $v$ , so it takes time at least  $d/v$ .

**Definition 3** (Measured observer). The observer  $\mathcal{O}$  is a device with parameters measured by the protocol of §5, in particular its solving time  $T_{\mathcal{O}}(n)$  for SAT instances of size  $n$ . The observer's HSAM [11, 12] is modelled as a large, accurate memory  $M_{\mathcal{O}}$  for the *problem statement*; it is not assumed to change the step time  $d/v$ .

## 4 Results

Let  $\text{SAT}_n$  be satisfiability of 3-CNF formulas on  $n$  variables; there are  $2^n$  assignments, each checkable in  $\text{poly}(n)$  steps.

**Theorem 1** (A: idealized collapse). *If  $v, E \rightarrow \infty$  and  $d \rightarrow 0$  with accelerating steps  $\tau_k = 2^{-k} s$ , all  $2^n$  assignments can be examined in total time  $\sum_k \tau_k \leq 1 s$ .*

*Derivation.* Examine assignment  $k$  during  $\tau_k$ ; the series converges to 1. This is a Zeno-type (accelerating) machine [21] and requires the speed, energy and step-size limits to be removed.  $\square$

**Theorem 2** (B: finite-parameter bound for brute force). *For a device with parameters  $(v, E, d)$  examining assignments sequentially,  $T(n) \geq 2^n d/v$ .*

*Derivation.* At least  $2^n$  sequential examinations are needed in the worst case, each at least one step of duration  $\geq d/v$ .  $\square$

**Theorem 3** (C: lookup bound). *A device that answers by table lookup can store answers for at most  $n_{\max} = \lfloor \log_2 M \rfloor$  variables.*

*Derivation.* A table of  $2^n$  one-bit answers needs  $2^n \leq M$ .  $\square$

*Remark 2.* For  $M \approx 8 \times 10^{15}$  bits (about 1 PB, an upper estimate of human neural storage [14]),  $n_{\max} \approx 52$ . Remembering the *formula* perfectly does not reduce the  $2^n$  candidates; it prevents losing the problem, not searching it.

**Proposition 4** (Erasure cost of brute force). *If each of the  $2^n$  examinations erases at least one bit at temperature  $T$ , the total heat is  $Q \geq 2^n k_B T \ln 2$ ; at  $T = 310 \text{ K}$ ,  $k_B T \ln 2 \approx 3 \times 10^{-21} \text{ J per bit}$ .*

*Derivation.* Multiply the Landauer bound [3] by  $2^n$ . Reversible implementations avoid erasure [5], so this bounds only irreversible search.  $\square$

## 5 Postulate and its status

**Postulate 2** (Correspondence principle). Computational collapse of NP is the limit of physical device behaviour as  $(v, E) \rightarrow \infty$ ,  $d \rightarrow 0$ ; at finite parameters, scaling is exponential for every physical device.

The first clause is Theorem A. Continuous-time analog SAT solvers show exponential trajectory-time growth on hard instances [20], and physical models of NP-hard computation are surveyed in [17]. NP-completeness is from [15]. For brute force the second is Theorem B. For *all* algorithms it would require a super-polynomial lower bound, which is  $P \neq NP$ , open. The postulate is therefore adopted, not derived, and is tested empirically (§5). The postulate mirrors the Newtonian limit of relativity: the idealized law is exact only in the limit.

## 6 Role of the observer in the argument

Three logical roles must be kept separate.

1. **Measured instrument.**  $T_{\mathcal{O}}(n)$  is data. It is compatible with the postulate if it grows exponentially, and refutes it if it is polynomial for large  $n$ .
2. **Not a second axiom.** That the observer “is the device” or that HSAM is “a rule of nature” is not used. HSAM is a rare documented trait. Its carriers show normal-range false-memory rates [13]; none of this makes it a universal law or a search accelerator.
3. **Not a proof.** One observer, however exceptional, cannot prove a statement about all inputs of all sizes. Finite data bound the scaling over  $n \leq n_{\max}$  only.

## 7 Experimental protocol

**Instances.** Random 3-SAT, clause/variable ratio  $\alpha = 4.27$  (the hardest region, near the satisfiability threshold [18, 19]), for  $n \in \{5, 10, 15, 20, 25, 30\}$ , at least 10 instances per  $n$ , generated by a third party or by a seeded program (Appendix) so the observer does not choose them. **Procedure.** The observer solves each formula without electronic aid and records the time  $T$  and the answer. **Scoring.** Each answer is verified: SAT answers by checking the assignment, UNSAT answers by a SAT solver. Report accuracy and median  $T$  per  $n$ . **Analysis.** Fit  $\log T = \log a + n \log b$  (exponential) and  $\log T = \log a' + k \log n$  (polynomial) on  $n = 5, \dots, 30$ ; compare  $R^2$  and information criteria. **Decision rule.**

- Exponential model preferred: consistent with the postulate; the slope gives the observer’s effective  $v/d$ .
- Polynomial model preferred over  $n \geq 20$  with accuracy  $\geq 95\%$ : refutes the postulate for this device and warrants replication with larger  $n$  and independent observers.

**Controls.** Rest between trials; no pencil, paper or screen; the same time limit per  $n$ ; blind instance order; the observer’s timing recorded by a separate person.

| $n$ | Instances | Correct | Median $T$ (s) | $\log_2 T$ |
|-----|-----------|---------|----------------|------------|
| 5   |           |         |                |            |
| 10  |           |         |                |            |
| 15  |           |         |                |            |
| 20  |           |         |                |            |
| 25  |           |         |                |            |
| 30  |           |         |                |            |

Table 1: Data sheet for  $T_O(n)$ .

## 8 Measurement and computation

The author’s argument is that information is in superposition and the observer’s measurement collapses it to a definite Boolean value. Measurement does yield a definite outcome, but by the Born rule it is random: for a uniform superposition over the  $2^n$  assignments of a formula with  $M$  satisfying ones,

$$p_{\text{success}} = \frac{M}{2^n},$$

which is  $2^{-n}$  for a unique solution. Grover’s algorithm raises this with  $O(\sqrt{2^n/M})$  iterations [48], and no quantum algorithm can search unstructured data with fewer than  $\Omega(2^{n/2})$  queries [44]. Macroscopic information is also not in superposition, since decoherence suppresses superpositions of macroscopic states rapidly [49].

**Specification of the missing procedure.** Closing the argument requires a family of quantum circuits  $\{C_n\}$  of size  $\text{poly}(n)$  such that, for every satisfiable  $\varphi$  on  $n$  variables,  $\Pr[C_n(\varphi) \text{ outputs a satisfying assignment}] \geq 2/3$ . This would give  $\text{NP} \subseteq \text{BQP}$ , which is not believed to hold. **No such circuit is given in this paper;** the procedure is recorded as an open requirement. *Under the author’s axiom*  $\text{P} = \text{NP}$  the requirement holds trivially, since  $\text{P} \subseteq \text{BQP}$ ; this assumes the conclusion and constructs nothing. The bound of [44] concerns unstructured black-box search and does not contradict  $\text{P} = \text{NP}$  (relativized worlds exist on both sides [16]), so it is not overridden by the axiom, and the author’s position that Grover’s and Zurek’s results “resolve themselves” under the axioms is recorded as a claim. The author’s claim that consciousness is solved is likewise recorded as a claim; the monograph’s qualia postulate is ill-posed (§1).

## 9 What general relativity adds

**It does not change the mathematics.** P vs NP concerns Turing machines and the zeros of  $\zeta$  concern complex analysis; both are independent of the spacetime in which one imagines computing. A proof of either is a mathematical text, not a physical effect [17].

**What it does allow, as model-relative collapses.** (a) Malament–Hogarth spacetimes contain an observer’s worldline of finite proper time that can receive the result of an unbounded computation run along another worldline; this gives non-Turing computation in the model [28]. Here the observer is used:  $\mathcal{O}$  selects the finite-length worldline. These spacetimes require special global structure and unbounded signal and energy resources, and their physical realizability is disputed [29]. (b) Closed timelike curves, in a suitable model, give  $P = PSPACE$  for classical and quantum computation [30]. Both results have the logical form of Theorem A: collapse relative to an idealized resource, not a proof that  $P = NP$  for Turing machines. Neither yields the Riemann hypothesis. The spectral approach [31] is an unproven conjecture.

**Open for a theory of everything.** Unifying general relativity with quantum mechanics (quantum gravity) is unsolved; the framework contains no such theory.

## 10 Routes, not solutions

Standard routes, with sources, for the six problems. None is claimed solved here.

- P vs NP: any proof must pass relativization [16], natural proofs [32] and algebrization [33]; survey [34].
- Riemann hypothesis: more than 40% of zeros on the critical line [35]; verified numerically to height about  $3 \times 10^{12}$  [36].
- Yang–Mills: constructive QFT, Osterwalder–Schrader reconstruction [37]; the 4D non-abelian case is open.
- Navier–Stokes: weak solutions [38], partial regularity [39], blow-up for an averaged equation [40].
- Hodge: only the degree-2 case (Lefschetz  $(1, 1)$ ) is proven.
- Birch–Swinnerton-Dyer: ranks 0 and 1 in many cases [41, 42].

The Poincaré conjecture was solved by Perelman [43].

## 11 Information and the Clay problems

The postulate of §2 supplies no bridge to any Clay problem. Information enters the literature of some of them as follows.

- P vs NP: query lower bounds are information-theoretic [44]; Kolmogorov complexity relates to complexity classes [45]; the erasure bound of §4 applies only to irreversible search [5].
- Yang–Mills: a spectral gap implies exponentially decaying correlations and, in one-dimensional gapped systems, an entanglement area law [46]. This is a consequence of a gap, not a route to proving one in four dimensions.
- Navier–Stokes: the relevant quantity is dissipation; for the Euler equations, Onsager’s conjecture was proved [47], which gives no regularity result.
- Riemann, Hodge, Birch–Swinnerton-Dyer: no information-theoretic route is known; for Riemann the nearest physics link is the unproven spectral conjecture [31].

All six remain open (Table 2).

## 12 Status of the other Millennium Problems

The exercise asked that the remaining Clay problems be treated as solved under the observer. A document cannot make a problem solved by declaring it so; what can be recorded is what each problem requires and that no solution is supplied here. Table 2 states this. The author states that these are solved; the table records them as open because no proof has been supplied for checking. Under the Clay Institute’s rules a solution must be published in a qualifying outlet and survive two years of general acceptance [22].

| Problem                    | Status here             | Required                                                                                                          |
|----------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| P vs NP [15]               | Open; no proof supplied | Polynomial algorithm for an NP-complete problem, or a super-polynomial lower bound; must pass relativization [16] |
| Riemann hypothesis [23]    | Open; no proof supplied | Proof that all non-trivial zeros of $\zeta$ have $\text{Re } s = \frac{1}{2}$                                     |
| Yang–Mills mass gap [24]   | Open; no proof supplied | Rigorous 4D QFT for a compact gauge group with spectral gap $\Delta > 0$                                          |
| Navier–Stokes [25]         | Open; no proof supplied | Global smoothness or finite-time blow-up in 3D                                                                    |
| Hodge conjecture [26]      | Open; no proof supplied | Hodge classes are rational combinations of algebraic cycles                                                       |
| Birch–Swinnerton-Dyer [27] | Open; no proof supplied | Rank = order of vanishing of $L(E, s)$ at $s = 1$                                                                 |

Table 2: Millennium Problems other than the Poincaré conjecture (solved by Perelman, 2003).

Any such solution would be a mathematical text, verified line by line by others, whatever the memory capacity of its author.

## 13 Conclusion

This work is a theory: its axioms are chosen, not discovered, and its consequences hold only relative to them. Proved: Theorems A–C. Adopted: the correspondence postulate. Not claimed:  $P = NP$  for Turing machines, or that HSAM yields a solution method. The observer’s contribution is a measurement, and the framework is falsifiable by that measurement.

## Appendix: instance generator

```
import random
def gen(n, seed, alpha=4.27):
    r = random.Random(seed); m = round(alpha*n); out = []
    for _ in range(m):
        vs = r.sample(range(1, n+1), 3)
        out.append([v if r.random() < .5 else -v for v in vs])
    return out    # clause = 3 signed literals, e.g. (1 OR -4 OR 7)
```

## Declarations

**AI assistance.** The formalization, audit and typesetting were produced with Claude (Anthropic); the framework and the observer role are the author’s. **Competing interests.** None declared. **Funding.** None. **Data availability.** No experimental data have been collected; the protocol and generator are in §7 and the Appendix.

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